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Modeling of the Calendering Process for Lithium-Ion Batteries
with DEM SimulationDavid Schreiner^{a,*}, André Klinger^a, Gunther Reinhart^a^a*Institute for Machine Tools and Industrial Management, Technical University of Munich, Boltzmannstr. 15, 85748 Garching, Germany** Corresponding author. Tel.: +49-89-289-15593; fax: +49-89-289-15555. E-mail address: david.schreiner@tum.de**Abstract**

The development of novel active materials and compositions in lithium-ion battery electrodes is a main research focus due to the increasing demand for electric mobility. Calendering, as the final step in electrode production, is a critical process that significantly influences the mechanical and electrochemical properties of the electrode. This paper will present an approach to predict the influence of calendering on the electrode material, composition, thickness and the machine behavior. Additional investigations provide guidelines for optimizing the process. All in all, a deep understanding of how the calendering process effects the electrodes is presented and forms the basis for further investigations.

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Keywords: lithium-ion battery, electrode production, calendering, process modeling, discrete-elemente methode (DEM)**1. Introduction**

Battery technology plays a crucial role to ensure climate goals in modern society [1]. The forthcoming electrification of powertrains, volatile power generation from renewable energies and a rapidly increasing number of mobile devices lead to the rising demand for energy storage technologies. High-performance, low-cost batteries have become a key feature for various applications. In electric vehicles, they appear today as a viable solution to minimize vehicular CO₂ and NO_x emissions [2]. As a consequence, the demand for batteries will increase substantially.

The properties of battery cells determine the benefit and acceptance of various products. Therefore, it is important to achieve a profound understanding of how the production process affects the battery cell and what special requirements the production processes of batteries have to fulfill. Not only the materials used but also the production processes determine the future performance of a lithium-ion cell [2].

The electrodes of lithium-ion cells consist of active material particles, binders and conductive additives, which are coated on

a metallic current collector foil. After the coating has been applied onto the collector foil, the solvent of the suspension evaporates in the subsequent drying process and forms a porous electrode structure [2]. Most of the state of the art cathode active material particles have a spherical particle structure in common [1] but differ in particle size and density. The state of the art anode active materials are carbonaceous materials and have either a flake-shaped or spherical particle structure [1]. Calendering faces the challenge of compacting a composite layer that is coated onto a metallic foil. With equal active material, the layer thickness of the coating significantly determines the ratio between specific power and specific energy [3]. In the application of an electric vehicle, the specific power influences the speed and acceleration, whereas the specific energy determines the range.

The production of lithium-ion batteries can be divided into two main categories: electrode production and cell assembly. The intricate processing with multiple parameters has a direct effect on the electrical, mechanical and electrochemical properties of an electrode [4]. The electrode production mainly

determines the properties of electrodes and thus significantly influences the performance of battery cells. One production step that has an impact on the subsequent production steps, as well as the product properties, is the calendering process. In the calendering process, the electrode material is specifically compacted through adjusting the desired porosity and therefore volumetric energy density of the electrode by applying a line pressure between two rolls. In the subsequent cell assembly, the electrodes, electrically insulated by a separator, are assembled and made into battery cells by means of contacting, electrolyte filling and formation.

Nomenclature

CABM	Conductive additive-binder matrix
CO ₂	Carbon dioxide
C65	Carbon black
DEM	Discrete element method
EEPA	Edinburgh Elasto-Plastic Adhesion
NO _x	Nitrogen oxide
NMC	Lithium-nickel-mangan-cobalt-oxide
PVDF	Polyvinylidene fluoride
SEM	Scanning electron microscope
SLS	Scattering light analysis
SFG6L	Timcal conductive graphite

2. State of the art of calendering

The mechanical properties of the electrode are determined by the substrate foil and the coating thickness [5]. Furthermore, the volume fractions of the materials, the particle size and shape [5] and the mixing process significantly influence these properties [2]. After coating and drying, there are pores within the structure due to the evaporated solvent, resulting in a low electrode density. Special attention is required for the compaction process [1] since energy density represents a very important property of battery cells. To increase the energy density, the electrodes are compressed to a higher volumetric density during the process of calendering [6]. The target electrode density is subject to a conflict between good ion conductivity (low density) and high electrical conductivity (high density) [7].

The mechanical homogenization by calendering [8] leads to homogeneous aging of the cell [3]. This can be explained by reduced inhomogeneity within the particle pore structure at higher densities [9]. Therefore, calendering is an important step to reduce variations in product quality, for example in relation to C-rate performance [3]. The C-rate is a value, which indicates the ratio of the discharge or charge current of battery cells relative to the nominal capacity. An analysis of the pore structure of electrodes through mercury porosimetry demonstrates that the compression of electrodes results in smaller pore sizes [10]. The denser particle structure can be explained by the mechanical stress applied during the compacting process, which causes rearrangements of the particles [11]. The porosity, adjusted by the calendering process, has a strong influence on the electrolyte filling and subsequent wetting time of the electrode [12].

Billot et al. [13] examined the adhesion strength of anodes and conclude that the key determinants for the adhesion strength of anodes are the binder content, the coating thickness and the calendering process. An increase in roll temperature influences the bond strength, as explained by the properties of the binder [13]. The shear modulus of the binder polyvinylidene fluoride (PVDF) decreases almost linearly with increasing temperature [14], which is the calender relevant temperature window up to 150 °C.

Sangros et al. [15] proposed a discrete element method (DEM) based approach to investigate the electrical and ionic conductivity as well as the adhesion strength of lithium-nickel-mangan-cobalt-oxide (LiNi_xCo_yMn_zO₂, with $x+y+z = 1$), commonly referred to as NMC, at different densification levels. The authors also consider numerically how the electrode structure changes during the intercalation and deintercalation of lithium-ions into the active material. Thus, Sangros et al. demonstrate the potential of simulations with the DEM to improve the understanding of the microstructure of lithium-ion electrodes [15].

Cathode-active material particles are harder than those of anodes. Therefore, higher forces must be applied to cause deformation in the form of particle-particle rearrangements, which are associated with particle friction [16, 17].

To our knowledge, however, the calendering process of electrodes and the effects on electrodes are scarcely addressed in the literature. In previous publications, the authors showed how the mechanical properties of the electrodes influence the production process [11] and what impact different process parameters have on the electrode, the machine and the calendering process [17]. The process modeling of the calendering process together with the prediction of calendering parameters has not been dealt with. So far, approaches to describe the microstructure of electrodes have been presented and the process of calendering was not deeply considered in these investigations [10, 15]. Since the calender rolls are in direct contact with the electrode and significantly define the properties of the electrodes, it is important that both, the calendering process and the microstructure can be optimally adjusted. The objective of this paper is, therefore, the introduction of a model, to support the selection of suitable calendering parameters. This results in a deeper comprehension and improvement of the calendering process. Additionally, the adjustment effort for calendering will be reduced, resulting in a significant decrease in the production costs of lithium-ion batteries during ramp-up.

3. Method overview

Different electrode materials and compositions, as well as varying cell requirements due to different applications, require an enhancement of the process design for the calendering. Therefore, this paper introduces a simulation-based process design, with which potential calendering parameters can be identified. The corresponding method includes experimentally pre-defined parameters and guidelines. An overview of the individual modules of the method is shown in Figure 1.

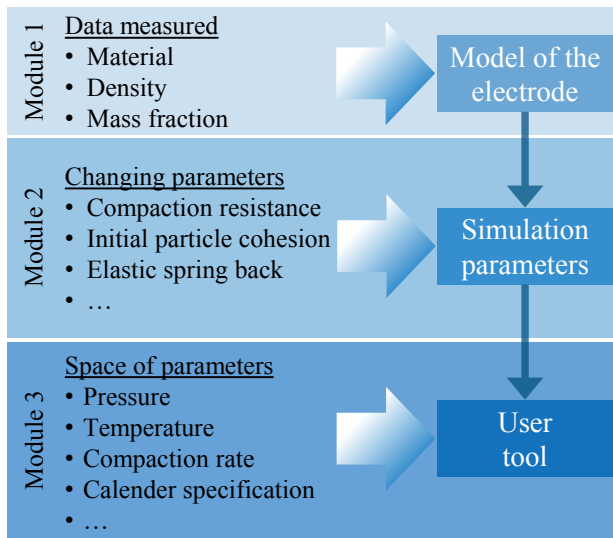


Fig. 1. Method overview for simulation-based process design of calendaring battery electrodes

The concept for calendaring process design encompasses three main modules: modeling of the electrode, parametrization of the electrode and identification of potential process parameters. First, the input and output product of the calendaring process, the electrode, is defined and then modeled. Based on the particle size distribution, density and thickness of the electrode, the control volume is defined and filled with particles. Verification of the electrode density and the particle distribution within the electrode concludes the model of the electrode.

Since the composition of the electrode is variable, the sensitive simulation parameters of the modeled electrode are parameterized in module two. A parameter screening is used to determine the sensitive parameters of the simulation. The parameters require calibration and are partly neither independent nor physically measurable, but merely represent adjustable parameters in a mathematical model. Therefore, the selection was made by microscopic analysis or preliminary simulative tests. The findings show that the numerical parameters can be categorized into four main categories: compaction resistance, elastic spring back, initial cohesion of the particles and others.

To reduce the time and effort required for calendaring settings, a tool for determining the calendaring parameters is developed. For this purpose, a calender roll section, which reproduces the diameter of the calender, is used. Subsequently, the required electrode properties are simulated with different calendaring parameters and suitable calendaring parameters are then suggested to the user. The user tool aims to provide the operator recommendations for suitable calendaring parameters.

4. Modeling of the electrode

Table 1 provides an overview of the material composition and densities of the individual components of the NMC622 cathode selected for the simulation.

Table 1. Mass fractions and density of the coating materials selected for the simulation

Description	Material	Crystal density in g/cm ³	Mass fraction in %
Cathode active material	NMC622	4.74	95.50
Binder	PVDF5130	1.70	2.25
Conductive carbon black	C65	1.66	1.50
Conductive graphite	SFG6L	2.33	0.75

The active material NMC622, with 95.5 wt%, represents the largest mass fraction of all components of the cathode. To calculate the theoretical volume of the electrode, the volumes of the individual electrode components (V_i) are determined. For this purpose, the measured mass is multiplied by the corresponding material content of the coating and divided by its density. Based on the values of Table 1 and measured single-sided coating thickness (d_{sheet}) of 56 μm , the coating density and porosity can be calculated according to equation (1) [18]. The density of the coating in the uncompressed state is 2.53 g/cm³, which corresponds to a porosity of 43 %.

$$\varepsilon_{\text{coating}} = 1 - \frac{\sum_{i=1}^n V_i}{A_{\text{sheet}} \cdot d_{\text{coating}}} \quad (1)$$

The particle size distribution was determined by static laser scattering light analysis (SLS). The measured particle size distribution of NMC622 particles has a logarithmic normal distribution with a mean particle size d_{50} of 7.19 μm . After analysis of the particle size distribution, the smallest particle in the simulation was defined as 3.4 μm , because the simulation time is decisively influenced by the smallest occurring particle. The volume fraction of the neglected particles is less than 2 % and has shown a negligible influence on the density of the electrode in preliminary tests. The upper limit was set to 20 μm in the simulation. Particles with a larger diameter would interfere with the production of the considered electrodes.

Figure 2 compares a scanning electron microscope (SEM) image with a DEM modeled electrode. Furthermore, Figure 2 c) shows a close-up of an NMC particle embedded in the binder and carbon black matrix, thus confirming that the assumption of spherical particles is qualitatively verified. The components binder and conductive additive appear as a conductive additive-binder matrix (CABM) between the active material particles. Neither a specific shape nor a representative particle size can be assigned to it. Consequently, it is assumed for the simulation that the conductive additives (C65 and SFG6L) and binder particles are not presented individually. Since the conductive additives are integrated into the conductive additive-binder-matrix or arrange themselves around NMC622 particles, the resulting deviation is negligible.

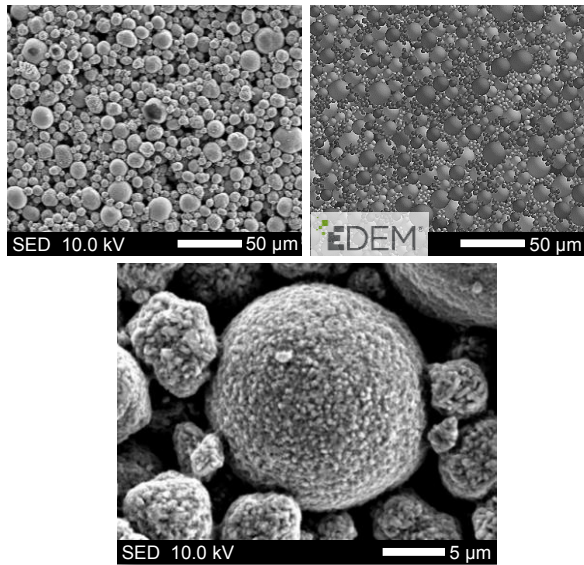


Fig. 2. a) SEM-image NMC622 b) Screenshot of DEM modeled NMC622 electrode, c) SEM-image 4000x zoom NMC622

The size of the control volume was determined as a square with an edge length of 250 µm. On the sides of the control volume, boundary conditions were assumed. The height of the control volume is set to 120 µm for the generation of the particles. The generation of the particles was implemented in a “dynamic factory”. The dynamic factory fills the control volume until it reaches the desired level by creating particles with a defined flow rate. In the next step, the electrode is trimmed to the considered electrode thickness, in this case 56 µm. This procedure has the benefit that the time-consuming particle generation process has no need for repetition if the composition remains constant whilst the layer thickness of the coating varies. Furthermore, trimming the surface layers avoids the usual edge effects of particle generation. To verify whether the particle bed of the electrode generated in the simulation corresponds to the real electrode, the densities of the electrodes were compared.

5. Parameterization of the compaction

The parameterization of the model is performed by pressure-displacement curves resulting from nanoindentation experiments. Figure 3 shows the pressure-displacement curve resulting from the nanoindentation of electrodes, as well as supporting schemes before (a), during (b) and after (c) the indentation.

A flat punch with a diameter of 100 µm and a constant indentation velocity of $0.15 \mu\text{m s}^{-1}$ was used as an indenter. The nanoindenter continuously logs the indentation depth into the sample and the applied force. When the specified penetration depth is reached, the nanoindenter is reversed with the same constant velocity of $0.15 \mu\text{m s}^{-1}$.

The Young's modulus of the particles has the greatest influence on the required compaction force, so this parameter was first investigated. If Young's modulus is increased, the required compressive force also rises. However, preliminary studies showed that this increase must be chosen carefully since Young's modulus has a direct influence on the Rayleigh waves. The simulation can lead to falsified results and the required

simulation time increases strongly with it. After preliminary adjustment of Young's modulus, the contact models are parameterized.

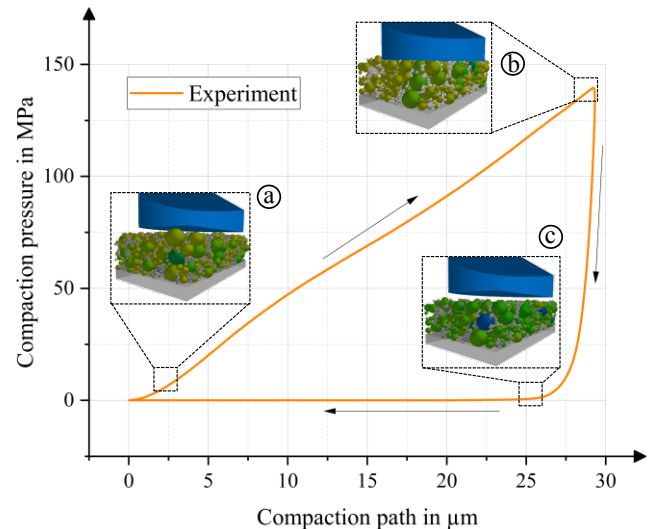


Fig. 3. Typical load-displacement curve for an NMC622 electrode before (a), during (b) and after (c) compaction.

The contact model for the interactions of the individual particles of the electrode was chosen because of the elastoplastic behavior of the electrode. The contact model describes the behavior of the electrode by interaction parameters between the particles. To reduce complexity, the parameterization was initially divided into two steps, with the two contact models: Edinburgh Elasto-Plastic Adhesion (EEPA) [19] and Bonding [20], being examined separately. Subsequently, the contact models were combined and a further parameterization was carried out.

The EEPA model was selected because, in addition to the elastoplastic behavior of the particles, the binding forces between the particles can be reproduced. These binding forces become particularly important with increasing compaction. The closer the particles get, the more pronounced the binding forces. Therefore, the EEPA is mainly responsible for the elastic spring back of the electrode after compaction [20].

Bonds are virtual contacts, generated at the first time step of the simulation. In the applied DEM software (EDEM®), the bond generation requires the bond contact radius as an input parameter. The contact radius is specified for each particle type and must be larger than the particle diameter. If the contact radii of two particles overlap, a bond is generated between them prior to the first simulation time step. After exceeding defined normal and shear forces, the bonds break. Once destroyed, bonds cannot be regenerated. Bonds primarily play a role at the beginning of compression, since an increase in the applied compression results in more bonds breaking. Therefore, the bonds have less influence with increasing compression levels. Bonding is used to enable a more pronounced increase in the pressure curve at the beginning. Bonds absorb force during compaction and break when the set limits are exceeded, resulting in a plastic behavior [20].

The parameterization of contact models is illustrated in the following using selected parameters for each contact model as illustrative examples. The contact plasticity ratio λ_p , defines

the unloading stiffness as a fixed ratio between the initial stiffness and the unloading stiffness. The contact plasticity ratio describes in the EEPA model the behavior of the particles in the unloading cycle after reaching the maximum compression. The value range of the contact plasticity ratio can be chosen between zero and one. The value zero represents a purely elastic and the value one a fully plastic behavior. Accordingly, an influential shift along the abscissa in the load displacement diagram is possible.

Figure 4 highlights individual influences when the contact plasticity ratio is increased in the different particle relationships. The compression path below 0 mm corresponds to the distance between the nanoindenter and the electrode. In the previous simulation illustrated in Figure 4 the contact plasticity ratio had values of $\lambda_{P/PVDF_NMC} = 0.9$, $\lambda_{P/NMC_NMC} = 0.8$ and $\lambda_{P/PVDF_PVDF} = 0.975$. Gradually these values were increased to 0.985. As shown in figure 4, the plastic interactions between NMC-Particles have the greatest influence in the EEPA, because the contact areas of these particles are larger than other particles.

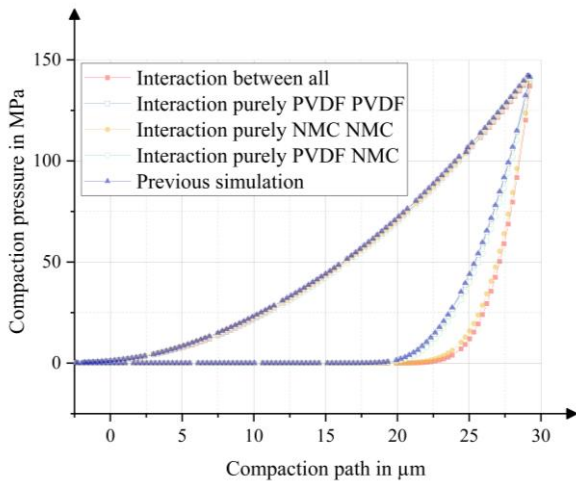


Fig. 4. Illustration of the influence of the contact plasticity ratio for purely NMC, purely PVDF and combinations of the two. The blue line represents the results of the previous simulation.

In the Bonding model, the choice of critical shear/normal stress τ_{Cr}/σ_{Cr} values for separating the bonds have to be selected. These values are used to directly control the resistance of the bonds. Although the simulation results presented in this paper only show a comparatively low influence on the simulation results, the parameters can play a decisive role in the final tuning of the simulation process. It is mentioned that critical shear/normal stress and Young's modulus correlate slightly indirectly.

Figure 5 shows the decreasing number of broken bonds across the compaction path and the load-displacement curve for the electrodes. As mentioned previously, Young's modulus, the contact plasticity ratio and the critical normal/shear stress values play a major role in the executed studies. It should be mentioned that some bonds are destroyed by the EEPA at the time step of bond generation ($t=0$ s).

After the impact and further translational compression of the compression plate to the control volume with the particles, the compression pressure increases. If the defined critical values are exceeded, bonds break during the compaction. At the values

depicted in figure 5, the intact amount of bonds decreases almost linearly in the compaction cycle and does not generate force jumps in the load-displacement curve. Through a progressive increase of each value, the experimental load-displacement curve can be more accurately reproduced.

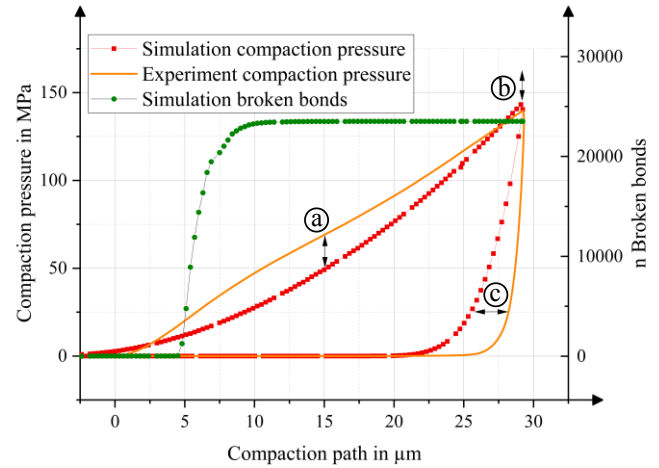


Fig. 5. Comparison of the load-displacement curves for the simulation and experiment, as well as the number of broken bonds. The shift during compaction (a) can be achieved through the change of critical shear/normal stress and surface energy. The magnitude of the compression pressure (b) is mostly defined by Young's modulus. The elastic recovery can mainly be determined by the contact plasticity ratio (c).

In addition, the surface energy $\Delta\gamma$ in the EEPA model was determined as an adjustable value for a load-dependent adhesion force f_{min} . Values used were: $\Delta\gamma_{NMC_NMC} = 5 J/m^2$, $\Delta\gamma_{PVDF_NMC} = 20 J/m^2$ and $\Delta\gamma_{PVDF_PVDF} = 60 J/m^2$. The required compression force of the compaction plate correlates with surface energy. For an approximation with the experimental values, only slight increases are recommended, otherwise, the required maximum compression force increases rapidly.

6. User tool

The aim of the user tool is to predict suitable process parameters for desired electrode properties and therefore reduce the adjustment effort of the necessary process parameters during calendering. Assuming that the elastic deformation of the rolls can be ignored, the theoretical roll gap corresponds to the smallest electrode thickness during the compaction process. As soon as the electrode passes the roller gap, the electrode springs back due to the recovery of the elastically deformed particles and trapped air in closed pores of the electrode. Since the phenomena described are dependent on the roll diameter and the compaction rate, a simulation of the roll geometry is necessary. With increasing roller diameter and decreasing compaction speed, the subsequent expansion is reduced due to better de-aeration and therefore a higher proportion of plastic deformation [21].

In order to determine the calender parameters, the roll diameter of the calender used is included in the simulation in the form of a roll section, as described previously. A calender roll section instead of the complete roll diameter is advantageous since otherwise, the simulation time increases.

This is due to the different order of magnitude between the particles ($\sim 10 \mu\text{m}$) and the roller ($\sim 400 \text{ mm}$).

Figure 6 provides an overview of the functionality of the user tool. The current and target properties of the electrode are input variables for the tool. The simulation is used to suggest possible calendaring parameters to achieve the target properties of the electrode.

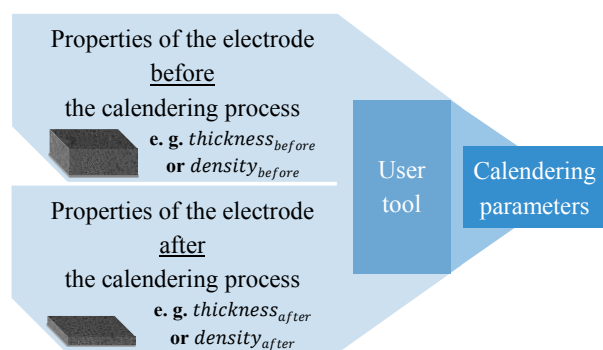


Fig. 6. User tool with illustration of input and output variables

For example, by setting the physical properties of the electrode and the information about the machine, the user is advised of potential calendaring parameters for setting the desired target property of the electrode. The parameters proposed by the model are based on the parameter areas examined by Schreiner et al. [17], as well as on literature and preliminary tests. The specifications of the target properties of the electrode, such as layer thickness, density or porosity, are determined by the requirements of the final battery cell and are taken as default values for the calendaring process.

7. Conclusion

Further enhanced battery technology is important to achieve improvements in the respective process steps. Within the scope of this paper, a modular three-step method is presented.

The method is intended to contribute a better understanding of the calendaring process of electrodes as well as an improved understanding of the effects of the individual process parameters. The method was tested and verified in a partial examination. Furthermore, sensitivity analyses of different parameters (Fig. 4 and Fig. 5) of the two selected contact models (EEPA and Bonding) were performed and presented. The combination of these models enables a good description of the behavior of the electrode. The modeled and parameterized electrode is used to determine suitable calendaring parameters as well as predict the effects of individual parameters.

Further research will focus on further improving the correlation between the results of the simulation with experimental studies. Additional investigations will address the question of how the temperature dependent compaction behavior of the electrode can be integrated. Furthermore, the electrodes calendared with different parameters will be investigated via mercury porosimetry to study their pore structure.

Moreover, future simulations will investigate further possibilities to reduce the simulation time. A possible method

could be the inertial scaling method, which increases the compression speed without distorting the simulation results.

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References

- [1] Schmuck, R., Wagner, R., Hörpel, G., Placke, T. et al., 2018. Performance and cost of materials for lithium-based rechargeable automotive batteries 3, p. 267.
- [2] Kwade, A., Haselrieder, W., Leithoff, R., Modlinger, A. et al., 2018. Current status and challenges for automotive battery production technologies 3, p. 290.
- [3] Lenze, G., Bockholt, H., Schilcher, C., Froböse, L. et al., 2018. Impacts of Variations in Manufacturing Parameters on Performance of Lithium-Ion-Batteries 165, A314-A322.
- [4] Haselrieder, W. *Kalandrierung zur gezielten Einstellung der Batterieelektroden-Performance*, 1st edn.
- [5] Günther, T., Billot, N., Schuster, J., Schnell, J. et al., 2016. The Manufacturing of Electrodes: Key Process for the Future Success of Lithium-Ion Batteries 1140, p. 304.
- [6] Wood, D.L., Li, J., Daniel, C., 2015. Prospects for reducing the processing cost of lithium ion batteries 275, p. 234.
- [7] Zheng, H., Tan, L., Liu, G., Song, X. et al., 2012. Calendaring effects on the physical and electrochemical properties of Li[Ni_{1/3}Mn_{1/3}Co_{1/3}]O₂ cathode 208, p. 52.
- [8] Haselrieder, W., Ivanov, S., Christen, D.K., Bockholt, H. et al., 2013. Impact of the Calendaring Process on the Interfacial Structure and the Related Electrochemical Performance of Secondary Lithium-Ion Batteries 50, p. 59.
- [9] Kang, H., Lim, C., Li, T., Fu, Y. et al., 2017. Geometric and Electrochemical Characteristics of LiNi_{1/3}Mn_{1/3}Co_{1/3}O₂ Electrode with Different Calendaring Conditions 232, p. 431.
- [10] Froboese, L., Titscher, P., Westphal, B., Haselrieder, W. et al., 2017. Mercury intrusion for ion- and conversion-based battery electrodes – Structure and diffusion coefficient determination 133, p. 102.
- [11] Günther, T., Schreiner, D., Metkar, A., Meyer, C. et al., 2019. Classification of Calendaring - Induced Electrode Defects and Their Influence on Subsequent Processes of Lithium - Ion Battery Production 5, p. 1900026.
- [12] Habedank, J.B., Günter, F.J., Billot, N., Gilles, R. et al., 2019. Rapid electrolyte wetting of lithium-ion batteries containing laser structured electrodes: in situ visualization by neutron radiography 102, p. 2769.
- [13] Billot, N., Günther, T., Schreiner, D., Stahl, R. et al., 2019. Investigation of the Adhesion Strength along the Electrode Manufacturing Process for Improved Lithium - Ion Anodes, p. 1801136.
- [14] Parry, E.J., Tabor, D., 1973. Effect of hydrostatic pressure and temperature on the mechanical loss properties of polymers: 2. Halogen polymers 14, p. 623.
- [15] Sangrós Giménez, C., Schilde, C., Froböse, L., Ivanov, S. et al., 2019.

- Mechanical, Electrical, and Ionic Behavior of Lithium - Ion Battery Electrodes via Discrete Element Method Simulations 217, p. 1900180.
- [16] Meyer, C., Bockholt, H., Haselrieder, W., Kwade, A., 2017. Characterization of the calendaring process for compaction of electrodes for lithium-ion batteries 249, p. 172.
- [17] Schreiner, D., Oguntke, M., Günther, T., Reinhart, G., 2019. Modelling of the Calendaring Process of NMC - 622 Cathodes in Battery Production Analyzing Machine/Material - Process - Structure Correlations 7, p. 1900840.
- [18] Gulbinska, M.K., 2014. *Lithium-ion Battery Materials and Engineering*. Springer, London.
- [19] Thakur, S.C., Morrissey, J.P., Sun, J., Chen, J.F. et al., 2014. Micromechanical analysis of cohesive granular materials using the discrete element method with an adhesive elasto-plastic contact model 16, p. 383.
- [20] Potyondy, D.O., Cundall, P.A., 2004. A bonded-particle model for rock 41, p. 1329.
- [21] Pietsch, W., 1987. *Roll pressing*, 2nd edn., London.